JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 85, NO. B5, PAGES 2594-2602, MAY 10, 1980

Paleomagnetism of the Triassic Hound Island Volcanics, Alexander Terrane, Southeastern Alaska

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The Hound Island Volcanics, a Late Triassic sequence of submarine basalts, has been investigated to determine possible large-scale movements of crust in southeastern Alaska. Data from oriented cores at 12 sites bordering Keku Strait yielded a mean paleomagnetic pole at latitude 23.0°N, longitude 188.8°E (α_{95} = 11.0°). Reliability of the pole is affirmed by a fold test and by the consistent antipolarity of normal and reversed magnetic directions. The anomalous position of this pole compared to that derived from coeval paleomagnetic data in central North America indicates that the region surrounding Keku Strait was rotated approximately 100° in a counterclockwise sense. The paleolatitude determined from the basalts is not significantly different from that predicted by data from the craton. Therefore the apparent northward drift of the Alexander terrane indicated by Paleozoic paleomagnetic results occurred before Late Triassic time.

INTRODUCTION

Paleomagnetic results from the Upper Triassic Hound Island Volcanics place important constraints on the timing of microplate movements in southeastern Alaska. The hypothesis that displaced microplates make up the westernmost belts of the northern Cordillera is founded on stratigraphic and biogeographic evidence [*Berg et al.*, 1972; *Jones et al.*, 1972; *Churkin and Eberlein*, 1977; *Monger*, 1977] and has been supported by recently obtained paleomagnetic evidence from Paleozoic rocks of the Alexander terrane [*Van der Voo et al.*, 1980]. The paleomagnetic results reported herein are from Triassic submarine basalt that is stratigraphically the uppermost part of the Alexander terrane on Kupreanof Island.

Most of southeastern Alaska lies within the Alexander terrane (Figure 1), a 100-km-wide belt extending from the Yukon-Alaska border to the south tip of Prince of Wales Island [*Berg et al.*, 1972, 1978]. The terrane consists of Ordovician through Triassic rocks that were deposited near a continental margin or possibly in an island arc. The Alexander terrane is separated from the west edge of the craton, defined as the Rocky Mountain trench, by several belts that include a belt of late Paleozoic oceanic crust. This oceanic terrane, the Cache Creek assemblage defined by *Monger* [1977], contains abundant chert, pillow basalt, and ultramafic rocks deposited in a deep ocean basin. Paleomagnetic evidence suggests that the Alexander terrane moved approximately 15° north sometime after the Carboniferous and thereby trapped the oceanic Cache Creek assemblage against the continent [*Van der Voo et al.*, 1980].

The geologic history of southeastern Alaska is complicated by the presence of an outermost belt, Wrangellia, part of which lies west of the Alexander terrane on Chichagof and Baranof islands [*Jones et al.*, 1977]. Wrangellia is characterized by a distinctive assemblage of Triassic tholeiitic basalt and limestone appearing in a narrow belt from Vancouver Island to the Wrangell Mountains in Alaska and includes the Queen Charlotte Islands and parts of southeastern Alaska (Figure 1). The anomalously low paleolatitudes from Triassic basalt of the Wrangell Mountains [*Hillhouse*, 1977*b*] and Vancouver Island [*Irving and Yole*, 1972] relative to those expected for central North America show that Wrangellia has moved several thousand kilometers north. Although the Triassic rocks of Chichagof and Baranof islands have not been sampled for paleomagnetism, we assume that this part of Wrangellia also drifted north.

Triassic basalt deposited over Paleozoic rocks of the Alexander terrane holds the key to the timing of microplate movements. We studied the Upper Triassic Hound Island Volcanics to answer the following questions: (1) did northward drift of the Alexander terrane occur after the Triassic?, and (2) were Wrangellia and the Alexander terrane linked together during this northward journey?

The paleomagnetic results presented here suggest that the Triassic rocks of the Alexander terrane have not moved north in relation to the craton. Therefore northward drift of the Alexander terrane occurred before the Late Triassic, and Wrangellia could not have been linked with the Alexander terrane during that movement. This result was expected on geologic grounds, as the Triassic sequences of Wrangellia and the Alexander terrane are too dissimilar in lithology to have been deposited close to one another [*Jones et al.*, 1977].

GEOLOGIC SETTING

The paleomagnetic sample sites (Figure 2) border the northwest end of Keku Strait, a passage separating Kuiu from Kupreanof Island. The geologic descriptions presented here are summarized from an earlier report by *Muffler* [1967], who set up the formal stratigraphic nomenclature and prepared a 1:63,360-scale geologic map of the region. We present a detailed summary of the pre-Triassic stratigraphy, because it links the basement to other parts of southeastern Alaska and delineates probable microplate boundaries in this tectonically complex region.

The oldest exposed rocks in the study area are Lower and Upper Silurian sandstone and argillite of the Bay of Pillars Formation. These complexly deformed turbidites are overlain by a thick brownish-gray limestone named the Kuiu Limestone of Late Silurian age. Marble, greenstone, and amphibolite of probable Devonian age crop out on the northwest tip of Kupreanof Island but do not occur on Kuiu Island. These rocks have been correlated with the Middle (?) Devonian Gambier Bay Formation of the southeast shore of Admiralty Island [*Loney*, 1964], approximately 50 km north of the Keku Islets. Thin-bedded tuffaceous argillite, volcanic graywacke,

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Paper number 80B0040.

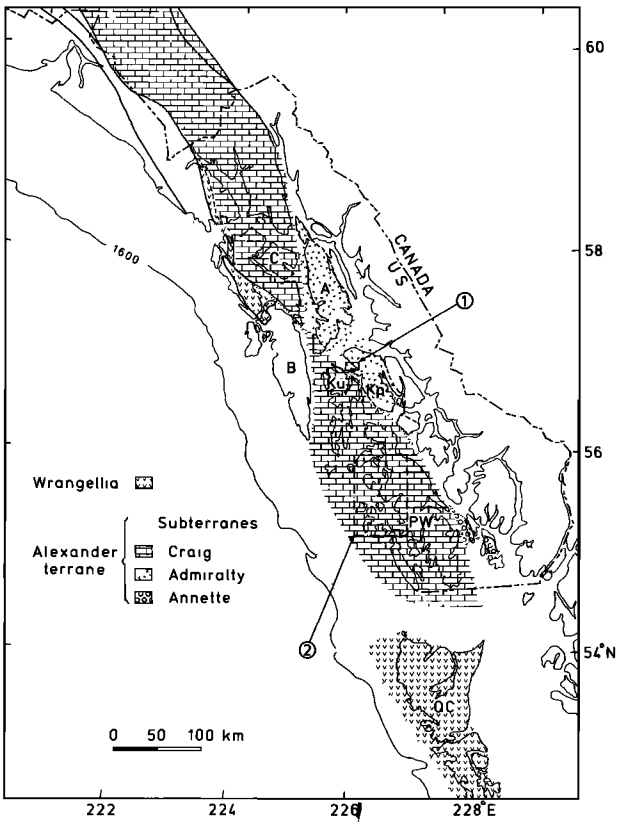


Fig. 1. Southeastern Alaska, showing study area (1) and location of Paleozoic rocks (2) studied by *Van der Voo et al.* [1980]. Terrane boundaries are according to *Berg et al.* [1978]. A is Admiralty Island; B, Baranof Island; C, Chichagof Island; Ku, Kuiu Island; Kp, Kupreanof Island; PW, Prince of Wales Island; and QC, Queen Charlotte Island.

and chert of Kupreanof Island are considered by *Muffler* [1967] to be part of the Cannery Formation, which was named by *Loney* [1964] for exposures on Admiralty Island. The Cannery Formation is considered to be of Early Permian age, but Late Devonian conodonts and radiolarians have recently been found in the chert of the Cannery [*Berg et al.*, 1978]. In addition, chert cobbles derived from the Cannery Formation were found in the lower Permian Halleck Formation, and these cobbles contain Late Devonian and Mississippian fossils.

The Paleozoic sequences of Kuiu, Kupreanof, and Admiralty islands are capped by the Permian Pybus Formation, composed of white limestone, dolomite, and chert. The Pybus Formation was deeply eroded before deposition of the Keku Volcanics during Late Triassic (Karnian) time. The predominantly felsic flows and breccia of the Keku Volcanics crop out on Cornwallis Peninsula of Kuiu Island, but the Karnian volcanic rocks do not extend to Kupreanof Island or Admiralty Island.

The samples in our study are from the Hound Island Volcanics, the uppermost formation of the Upper Triassic Hyd Group. This group consists of a basal conglomerate overlain by limestone and spilitic volcanic rocks. The Hyd Group lies conformably on the Keku Volcanics of Kuiu Island but on Kupreanof Island was deposited directly on the eroded surface of the Pybus Formation. Equivalent rocks on Admiralty Island are called the Hyd Formation [*Loney*, 1964].

Near Hamilton Bay the Hound Island Volcanics are overlain by lithic sandstone and mudstone of Early Cretaceous age; in turn, these beds are overlain by nonmarine sedimentary rocks of probable Paleocene age. Large gabbro bodies and associated microgabbro dikes intrude the early Tertiary sedimentary and older rocks. Though not dated directly, the gabbro is probably correlative with Eocene and(or) Oligocene basalt on Admiralty Island to the north. Gabbroic intrusive rocks occur in the Hound Island Volcanics at the north end of Hamilton Island and on the northern Keku Islets.

As was revealed by a recent study [*Berg et al.*, 1978] in southeastern Alaska, the Paleozoic rocks of the Alexander terrane can be split into three subterranes, each characterized by distinctive stratigraphic relations. The largest subterrane, the Craig, consists of the rocks on Prince of Wales Island, Kuiu Island, Chichagof Island north of Peril Strait, and the Glacier Bay region (Figure 1). The Craig block has been severed by right-lateral displacement of approximately 100 km on the Chatham Strait fault [*Brew et al.*, 1966] since Triassic time. The Cannery and Gambier Bay formations on Kupreanof Island and Admiralty Island define a second subterrane. These rocks differ from Devonian rocks of the Craig subterrane. The two subterranes were sutured together by Permian time because the Pybus Formation overlies both. Annette Island and part of Gravina Island make up the third subterrane. By Late Triassic time, all three blocks were sutured together, as is indicated by a mantle of equivalent Triassic formations—the Nehenta and Chapin Peak formations and the Hyd Group, which includes the Hound Island Volcanics.

The Hound Island Volcanics consists of basaltic pillow lava and pillow breccia, andesitic breccia, aquagene tuff, and minor limestone lenses. The unit is about 650 m thick at the north tip of Kupreanof Island and thins to about 100 m on Kuiu Island. The dominant rock type is broken pillow breccia, but pillow and massive lava flows are also abundant. Massive lava flows, ranging from 0.5 to 15 m in thickness, are well exposed on Hamilton Island. Pillow flows are best exposed on the shore of Hound Island and on the north shore of Hamilton Bay. The pillows are generally small, about 20 cm in diameter; radial joints, concentric bands of vesicles, and chilled margins are well-preserved features of the pillows. A detailed geologic map of Hound Island has been published by *Brew and Muffler* [1965].

As described by *Muffler* [1967, pp. C36–C37], the basaltic pillow lavas '. . . . are composed of microphenocrysts of labradorite and titanaugite set in devitrified groundmass consisting of cryptocrystalline chlorite, clay, opaque material, and sporadic microcrystalline epidote.' Also, 'the massive basalts are fine grained, as contrasted to the microcrystalline texture of the pillow lavas and breccias, and contain labradorite and titanaugite in chlorite-plagioclase mesostasis' [*Muffler*, 1967, p. C36]. Invertebrate fossils (Tropites, Halobia, and Monotis) from limestone interbedded with the volcanic rocks are of latest Karnian to late Norian age and securely place the Hound Island Volcanics in the Late Triassic.

The Hound Island Volcanics of Hound Island is rather remarkable, because the bedded tuffs and volcanic sandstone contain abundant fragments of fresh basaltic glass, one of the very few documented occurrences of pre-Tertiary undevitrified glass in the world [*Brew and Muffler*, 1965; *Muffler et al.*, 1969]. This fact led us to expect a high probability that the paleomagnetic record would also be well preserved in these rocks.

PALEOMAGNETIC SAMPLING AND LABORATORY TECHNIQUES

We sampled exposures of the Hound Island Volcanics on the shorelines of Hamilton Island and Hound Island in the

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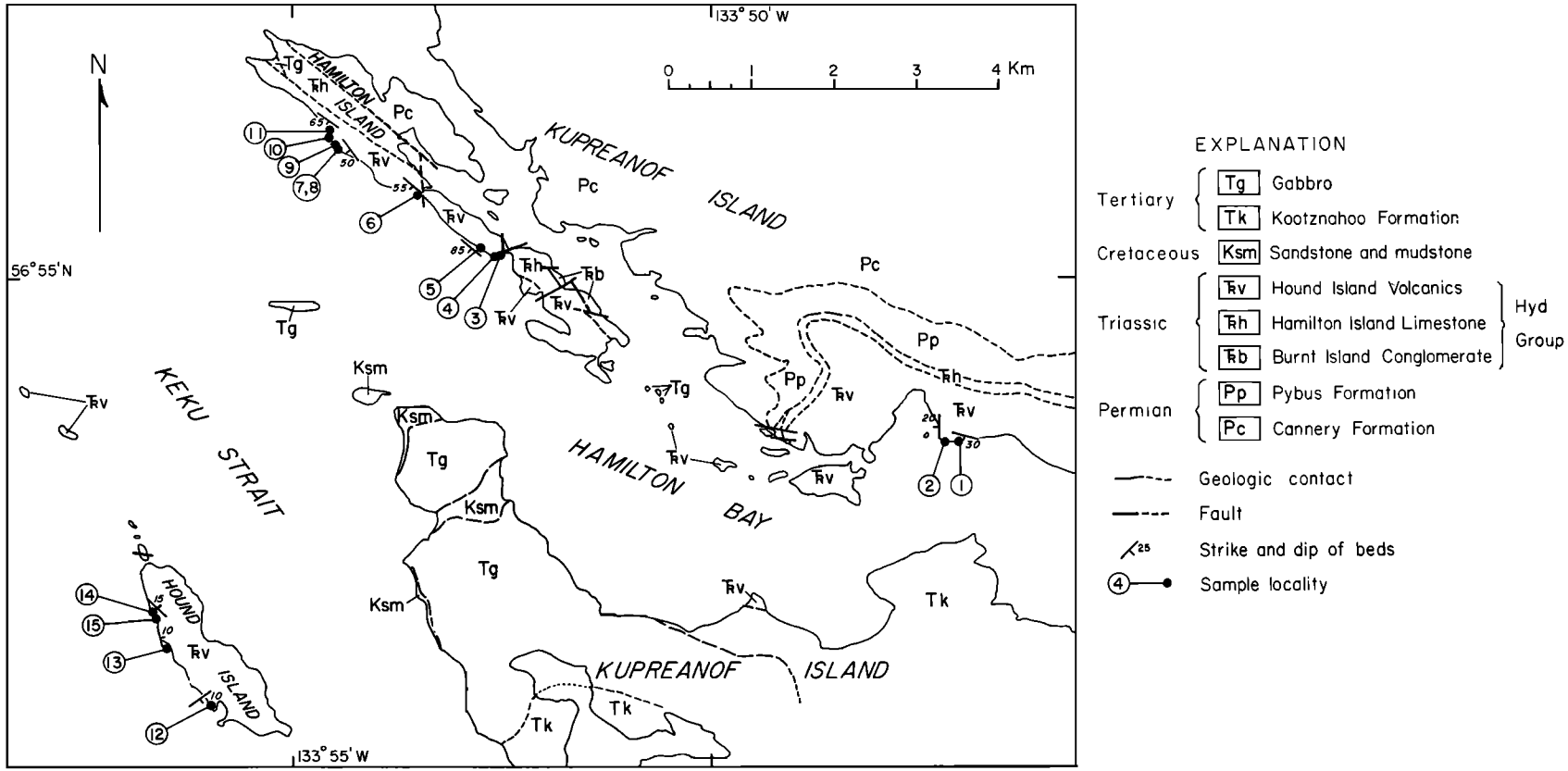


Fig. 2. Geologic map of northern Keku Strait area, showing locations of paleomagnetic sample sites.

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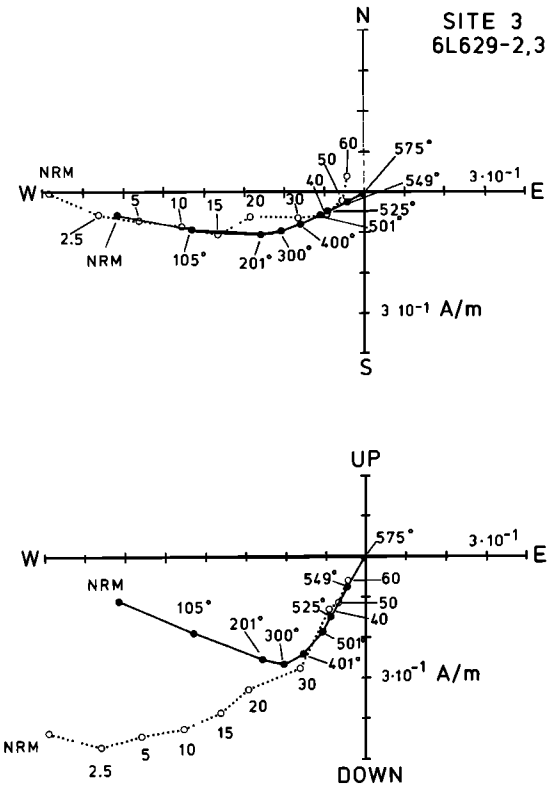


Fig. 3*a*

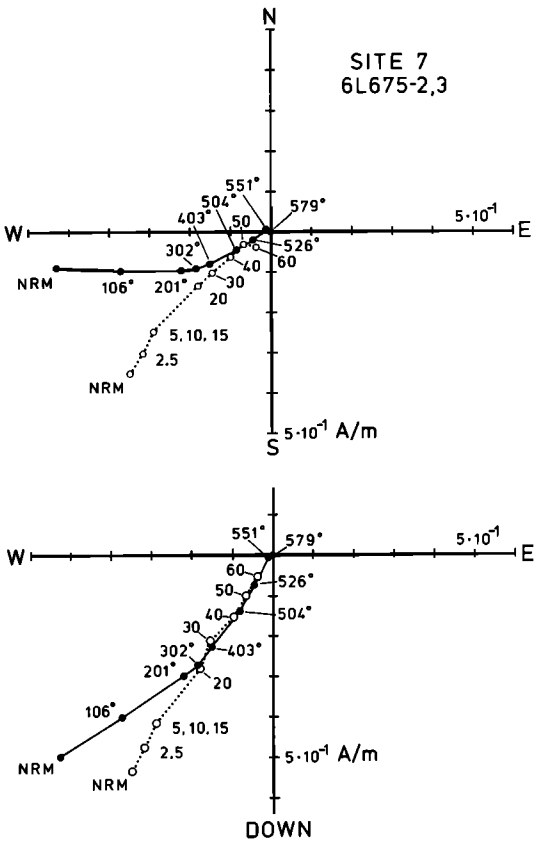


Fig. 3*b*

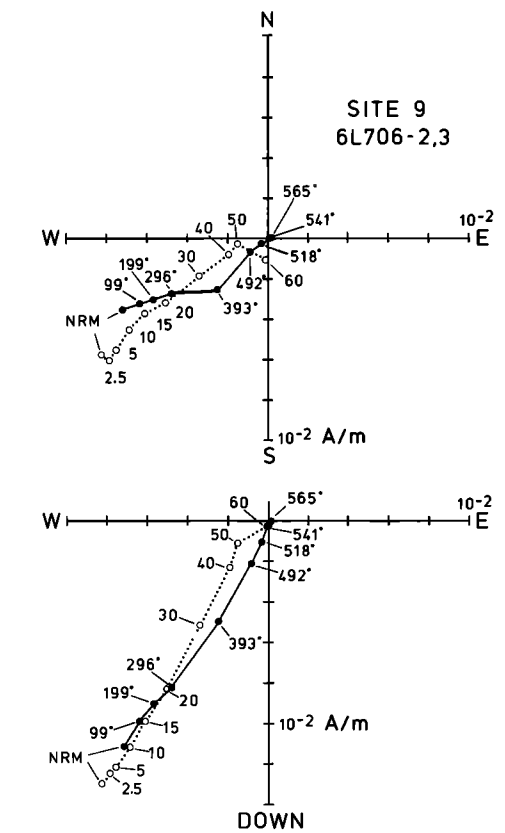


Fig. 3*c*

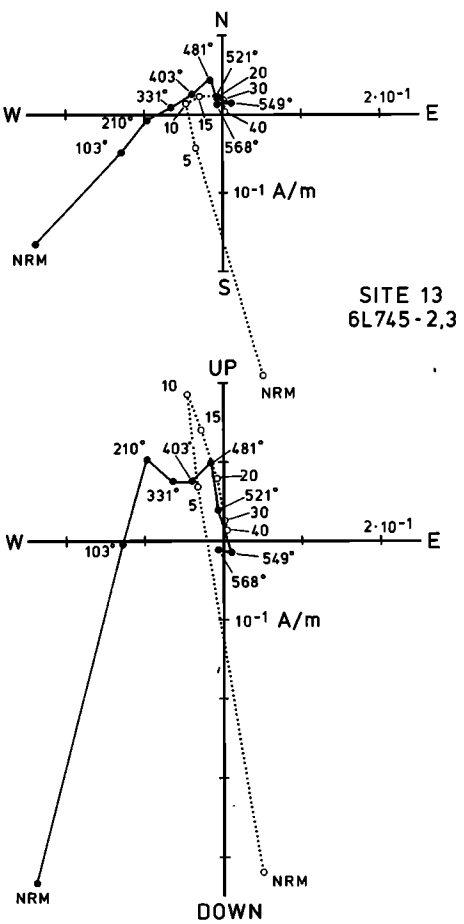


Fig. 3*d*

Fig. 3. Representative vector diagrams for sample sites (*a*) 3, (*b*) 7, (*c*) 9, and (*d*) 13 from stepwise alternating field (open circles) and thermal (solid circles) demagnetization. Secondary components usually vanished in AF's of 30 mT or at temperatures of >500°C. The example from site 9 retained multicomponent remanence during treatments in AF's as strong as 50 mT, a feature shown by ~25% of pilot specimens.

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Petersburg (D-6) quadrangle (Figure 2). A total of 175 cores were drilled from massive basalt flows and pillow basalt at 15 sites (see Figure 2 for locations), each site being a single flow unit represented by 8–18 cores (2.5 cm in diameter). Orientations were determined with a magnetic compass and an inclinometer.

Two independent paleomagnetic investigations were made on companion specimens cut from each core. The first study, begun in 1966, entailed measurement of natural remanent magnetization (NRM) and partial demagnetization of the specimens in alternating fields (AF's) limited to 30 mT. Higher demagnetizing fields were not applied because the available spinner magnetometer lacked the required sensitivity, so that these data have been omitted from Table 1.

A second subset of untreated specimens was studied in 1977, using a more sensitive spinner magnetometer and a cryogenic magnetometer. In this study, AF's as strong as 60 mT were applied to pilot specimens in order to test further the stability of remanence. When the dispersions of magnetic directions from these two studies were compared, a considerable improvement was seen in the second study, owing to the application of stronger cleaning fields allowed by the more sensitive magnetometers.

All demagnetization treatments were made in an unshielded four-axis tumbler, similar to the device described by *Doell and Cox* [1967]. Each demagnetization step was repeated with the specimen inverted in the tumbler to check reproducibility of the treatment and to remove spurious rotational remanent magnetization (RRM) [*Hillhouse*, 1977*a*]. Appreciable RRM was acquired by some specimens after treatment in fields exceeding 40 mT.

Curie temperatures of representative specimens were measured with an automatic thermobalance that determined strong-field magnetization as a function of temperature. Specimens were heated to 620°C in steady fields of 200–800 mT; heating and cooling rates were 10°C/min. A few heatings attempted in vacuum were unsuccessful because gases streaming from decomposing secondary minerals disturbed the balance; thereafter all specimens were heated in nitrogen gas. Thermal demagnetization was performed in a weak-field (50 nT) furnace maintained at $\sim 10^{-5}$ torr.

Paleomagnetic Results

Two pilot specimens from each sample site were treated in progressively stronger AF's ranging from 2.5 to 60 mT. Stability of the remanence during progressive demagnetization was determined from vector diagrams [*Zijderveld*, 1967], which generally revealed two and sometimes three superimposed components of magnetization. Vector subtraction indicated that the least stable component, which was removed by an AF of 20–30 mT, paralleled the recent geomagnetic field at the site. In AF's stronger than 20–30 mT, most specimens exhibited a single characteristic magnetization indicated by linear decay of the magnetic vector toward the origin of the diagram (Figures 3*a*, 3*b*, and 3*d*). The specimens usually acquired spurious remanence due to instrumental noise in AF's exceeding 50 mT, at which the remanence had been reduced to less than 5% of the NRM.

After treatment in an AF of 60 mT, approximately 25% of the pilot specimens had vector components that failed to converge linearly to the origin of the vector diagram (Figure 3*c*). Apparently, the specimens possessed a highly coercive component of magnetization characteristic of hematite, in addition to the components previously mentioned. These persistent magnetizations, making up 5–10% of the NRM, were generally along the recent geomagnetic field direction and were probably due to authigenic hematite.

Demagnetization of the remaining specimens was carried out at two to four field steps selected from the representative vector diagrams. The best estimate of the characteristic magnetization was determined from the cleaning step which maximized k, the Fisher precision parameter at each site. AF treatment substantially improved k in comparison with the corresponding values for NRM at all but two sites.

The paleomagnetic directions obtained from the optimum demagnetization step were then selected so as to minimize random errors possibly introduced by the experimental procedure or by natural remagnetization. We rejected specimens on the following basis: all directions at a sample site were used initially to calculate a mean and angular standard deviation, and then any directions lying more than two angular standard deviations from the mean were rejected. This procedure, which was used only once before a new mean and statistic were calculated (Table 1), resulted in the rejection of three specimens.

Thirteen representative specimens were thermally demagnetized. These specimens were cut from the same cores used in the pilot AF demagnetization studies, the strong-field thermomagnetic experiments, and the polished-section observations. Vector diagrams obtained from the thermal demagnetization runs were similar to corresponding diagrams from the pilot AF demagnetization runs (Figure 3). Many thermal vector diagrams showed two components of magnetization: a low blocking temperature component J_A that vanished above 400° and a high-temperature component J_B that was revealed between 400° and 550°C. J_A was similar to the components removed in weak alternating AF's and was generally directed along the recent geomagnetic field direction. J_B was nearly identical to the single high-coercivity component isolated in many of the pilot AF-thermal specimens. However, the highly coercive contamination observed in approximately 25% of the AF vector diagrams was not apparent in the thermal experiments. Heating to the Curie temperature of magnetite (580°C) removed virtually all the measurable remanent magnetization.

Thermal demagnetization appeared to be more effective in removing the viscous or chemical secondary remanence that affected several pilot specimens. However, AF demagnetization in the range 20–40 mT yielded nearly identical characteristic directions despite its failure to remove the small secondary component completely. We concluded that thermal treatment of the entire collection was unnecessary.

The structural setting of the sample sites (Figure 2) provides a good test of whether the remanent magnetization predates the deformation that occurred probably during the Late Cretaceous. The sites are on both limbs of the Keku synclinorium, a broad northwest trending structure centered in Keku Strait; the plunge of the structure appears to be negligible. Strata sampled from the shore of Hamilton Island dip 40°–65° southwest; strata on the west shore of Hound Island dip 10°–20° northeast. Attitudes of the volcanic rocks are well controlled at most sites by interbedded sedimentary rocks or by planar-flow structures. However, poor exposures at three sites (sites 1, 2, and 12 in Figure 2) where local deformation is intense made the bedding corrections uncertain. In view of this uncertainty and the wide disagreement between the data from these three sites and the remaining data we omitted sites 1, 2, and 12 from further analysis.

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TABLE 1. Summary of Paleomagnetic Results From the Hound Island Volcanics, Southeastern Alaska

Site	Latitude, °N	Longitude, °E	N	J, A/m	I°	D°	Ic°	Dc°	Polarity	R	k	α_{95}	Θ°	ϕ°	H, mT
1*	56.88	226.22	12	2.41, −2	66.0	41.7	76.6	143.0	N	11.9275	152	3.5	34.7	244.5	15
2*	56.88	226.21	10	6.10, −3	39.0	70.3	57.3	61.0	N	9.4731	17	12.0	46.6	314.9	20
3	56.92	226.13	11	2.26, −1	56.7	25.8	55.3	237.6	N	10.9216	128	4.1	14.6	181.0	40
4	56.92	226.12	10	1.26, −1	56.4	23.9	55.0	239.5	N	9.7357	34	8.4	15.2	179.5	30
5	56.92	226.12	18	4.21, −3	48.9	28.5	62.9	241.7	N	17.9749	678	1.3	23.6	182.8	20
6	56.93	226.11	13	5.52, −3	52.6	50.8	72.2	243.4	N	12.6865	38	6.8	34.9	190.0	30
7	56.93	226.09	10	9.88, −2	65.4	21.0	45.0	237.3	N	9.8615	65	6.0	6.4	176.9	30
8	56.93	226.09	13	2.78, −3	52.9	35.5	59.4	234.1	N	12.9102	134	3.6	17.3	185.7	20
9	56.93	226.09	10	2.86, −3	64.6	52.5	48.0	218.6	N	9.6622	27	9.5	2.0	193.0	15
10	56.93	226.09	10	2.17, −1	61.1	31.7	68.5	228.7	N	9.7781	41	7.7	25.8	195.0	20
11	56.93	226.09	10	9.80, −2	58.7	22.7	52.2	247.4	N	9.8886	81	5.4	16.1	172.2	40
12*	56.87	226.07	11	8.11, −2	45.8	332.2	48.6	344.8	N	10.6939	33	8.1	60.7	73.8	20
13	56.88	226.06	10	7.48, −2	−71.7	58.8	−77.6	341.3	R	9.5157	19	11.5	33.9	235.0	20
14	56.88	226.06	9	5.06, −2	−51.8	43.3	−66.5	54.2	R	7.9845	8	19.6	25.0	190.1	15
15	56.88	226.06	15	1.20, −1	−69.9	45.9	−82.4	92.8	R	14.7708	61	4.9	54.7	199.5	20

The overall mean VGP parameters are $\Theta = 23.05°$, $\phi = 188.80°$, $\alpha_{95} = 11.0°$, $K = 17$, and $N = 12$ sites. N is the number of specimens; $J = a$, $-b = a \times 10^{-b}$ is the mean intensity of cleaned remanence; I and D are the inclination and declination without bedding tilt corrections; Ic and Dc are the inclination and declination corrected for tilt of bedding; N and R are normal and reversed polarities; R is the resultant vector; k is the Fisher precision parameter for directions; α_{95} is the 95% confidence radius; Θ is the latitude of VGP; ϕ is the longitude (east) of VGP; H is the alternating field treatment.

*Omitted from calculation of overall mean VGP, because bedding corrections were uncertain.

Unfolding the Keku synclinorium clearly reduces the scatter of site directions by converging the directions from Hamilton and Hound islands in the southwestern quadrant (Figures 4*a* and 4*b*). The Fisher precision parameter k for directions increased from 9 to 33 after application of the bedding correction. This reduction of directional scatter is significant at the 99% level, according to the F test developed by *Watson and Irving* [1957], and indicates that the magnetization predates the folding.

As is shown in the tilt-corrected results of Figure 4*b*, reversed inclinations tend to be steeper than the normal ones, so the overall distribution of directions is elliptical. This departure from the expected circular distribution may indicate incomplete sampling of secular variation in the reversed polarity zone or may indicate the incomplete removal of secondary components. We consider any bias introduced by these sources of error to be negligible, because the normal and reversed directions are antipolar within the 95% confidence limits. Antipolarity of the data, which indicates a dipolar field, rules out the possibility that a short-period geomagnetic excursion was sampled. By assigning equal statistical weight to each site we calculate a mean paleomagnetic pole at latitude 23.0°N, longitude 188.8°E (α_{95} = 11.0°) for the Hound Island Volcanics.

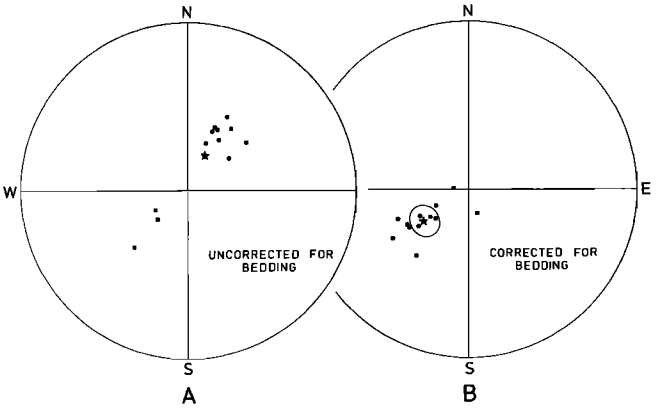


Fig. 4. Sample site mean magnetic directions: (*a*) in situ and (*b*) after correction for tilt of bedding. Circles denote normal polarity directions in the lower hemisphere; squares denote reversed directions that have been inverted to the lower hemisphere. The mean direction (star) is shown with its α_{95}.

Rock Magnetism

Polished thin sections examined at 1050× in oil revealed complex assemblages of magnetic minerals. The massive basalt flows contain iron-titanium oxides that exhibit two stages of oxidation: a primary deuteric alteration acquired at high temperature and a presumably later stage of maghemitization acquired at low temperature. (Magnetic grains in the pillow basalts were too small to be observed at 1050×.) Most magnetic grains consisted of two-phase intergrowths of titanomagnetite and ilmenite lamellae, the products of equilibrium at high oxygen fugacity and high temperature. However, deuteric alteration was incomplete in most specimens, as is shown by the coexistence of homogeneous titanomagnetite grains with two-phase grains. Rods of homogeneous ilmenite were also observed in a few polished sections. Partial maghemitization was common around shrinkage cracks and corrosion pits in the titanomagnetite grains. In several sections, ilmenite lamellae were partly replaced by a nonreflecting mineral that we assume to be sphene, a common product of low-temperature burial metamorphism [*Kuniyoshi and Liou*, 1976]. Reddish haloes due to the formation of limonite were rare.

A representative thermomagnetic curve was determined for each sample site, using the same cores made into polished sections. The curves fall into two types, designated strong (S) or weak (W), depending on the initial value of strong-field magnetization and the overall shape of the curve (Figure 5). W-type curves have a very low strong-field magnetization during heating and poorly defined Curie temperatures ranging from 530° to 580°C. The strong-field magnetization of the W-type curves markedly increased during cooling, although the Curie temperatures tended to decrease. All specimens of pillow basalt gave W-type thermomagnetic curves. S-type curves started with high strong-field magnetizations and exhibited well-defined Curie temperatures of 510°–565°C during heating. Curie temperatures were generally lower by 10°–50°C

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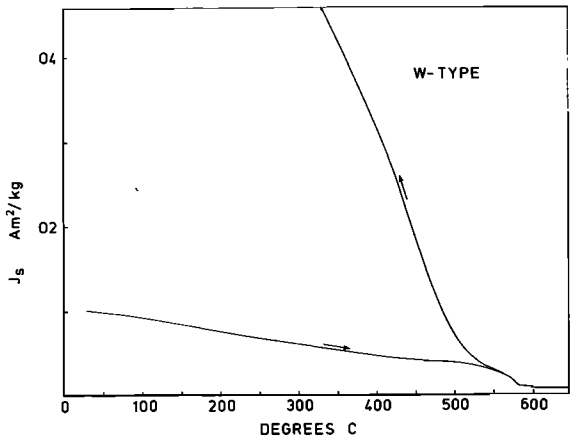


Fig. 5a

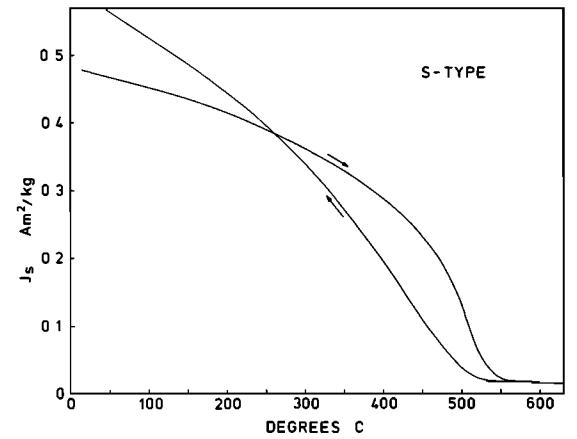


Fig. 5b

Fig. 5. Representative thermomagnetic curves showing two types of behavior commonly observed when samples of Hound Island Volcanics were heated in nitrogen: (*a*) W-type curve for massive basalt from site 6; strong-field magnetization was measured in a 400-mT field. (*b*) S-type curve for massive basalt from site 9; 400-mT field (1 Am^2/kg = 1 emu/g).

during the cooling cycles, but the final strong-field magnetization was equal to or slightly greater than the initial value. Most massive lava flows, with a few exceptions, exhibited S-type behavior.

The initial Curie temperatures of all specimens were somewhat lower than 580°C, the Curie temperature of pure magnetite, and indicate the presence of titanium-poor titanomagnetite. This result is consistent with our microscopic observation of deuterically altered two-phase grains.

Th magnetic minerals in all specimens were chemically altered during the heating experiments, as is indicated by the irreversible thermomagnetic curves. After the specimens were heated above 560°C, the Curie temperatures were lowered by 10°–50°C, and the strong-field magnetizations measured at room temperature tended to increase. In addition, susceptibilities tended to increase during the progressive heating steps of the thermal demagnetization experiments. These changes in magnetic properties were accompanied by sharp weight losses when specimens were heated above 525°C; we assume that these losses were due to the breakdown of clay minerals, chlorite, and sulfides.

We assume that the stable remanence detected in these basalts resides in titanomagnetite lamellae formed during original cooling. Apparently, the basalt flows were erupted into a shallow marine environment [*Muffler et al.*, 1969] in which the oxygen fugacity was high enough to allow deuteric formation of two-phase iron-titanium oxides at high temperatures. Mild heating and chemical changes during subsequent burial metamorphism failed to erase the primary magnetization carried by the finely divided titanomagnetite lamellae. Our claim of primary remanence is substantiated by positive results from the fold test and the consistency of normal and reversed directions.

DISCUSSION

If we are justified in assuming that the mean paleomagnetic pole (latitude 23.0°N, longitude 188.8°E; α_{95} = 11.0°) represents the Late Triassic geocentric axial dipole field, then we can infer the tectonic history of southeastern Alaska. Justification of this assumption requires that the sample sites span enough time to average out geomagnetic secular variation and excursions. We have sampled 12 different basalt flows, each providing a spot reading of the ancient geomagnetic field. Although the exact time interval spanned by these 12 flows is unknown, the flows were erupted during at least two polarity intervals. Consistent dual polarity of the data strongly indicates dipolar behavior that probably is not biased by secular variation or an excursion. Therefore the paleomagnetic pole from the Hound Island Volcanics adequately represents the position of the ancient geographic pole (Figure 6).

A mean Late Triassic reference pole (latitude 65.3°N, longitude 94.2°E; α_{95} = 5.9°) for central North America is established by data from six volcanic units near the Atlantic coast and from red beds in Utah. The various poles are identified as I8.44, I8.62-64, I8.111, I8.213, and I8.237 in the paleomagnetic catalogs compiled by E. Irving and co-workers [*Hicken et al.*, 1972; *Irving et al.*, 1976]. An equal-area map (Figure 7) centered on Keku Strait shows that the pole for the Hound Island Volcanics is displaced counterclockwise from the cluster of reference poles. However, the paleomagnetic colatitude inter-

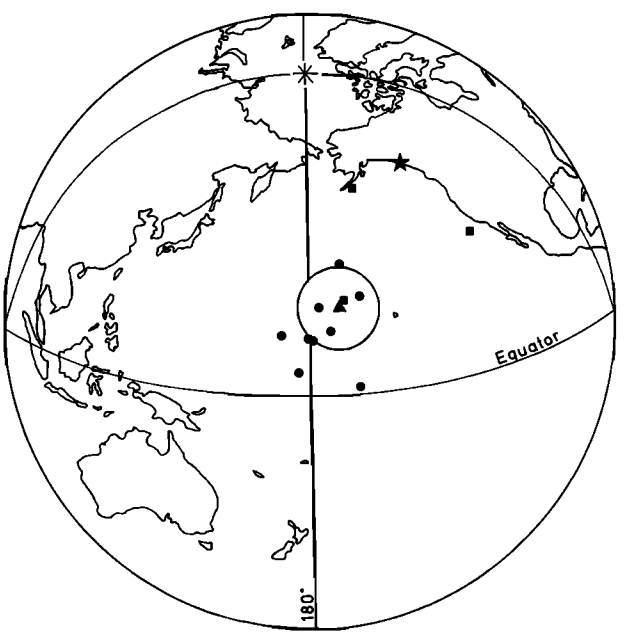


Fig. 6. Virtual geomagnetic poles (circles represent normal polarity; squares, reversed sites transposed), mean pole (triangle), and α_{95} determined from data at 12 sample sites within Hound Island Volcanics. The star indicates the study area.

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Fig. 7. Equal-area map centered on the study area (star in circle), showing the distribution of Late Triassic virtual geomagnetic poles (circles) from sample sites in central North America (squares) and the paleomagnetic pole (triangle) determined from Hound Island Volcanics. The mean Late Triassic paleomagnetic pole for central North America is shown by the star encircled by its corresponding α_{95}.

preted from this pole does not differ significantly from the group of colatitudes predicted from the reference poles. Therefore post-Triassic northward drift of the study area relative to the craton is unlikely. Of course, the region may have moved east toward the continent along a small circle centered on the mean reference pole, but this hypothesis cannot be tested without a more complete polar wander curve.

The counterclockwise rotation (100° ± 16°) inferred from the pole for the Hound Island Volcanics is in the same sense as the rotation suggested by *Van der Voo et al.* [1980] for the Paleozoic rocks near Craig, Alaska, although the Craig rotation (25°) is smaller. In view of the structural complexity of southeastern Alaska we conclude that these rotations are valid only for the limited areas surrounding the sample sites and do not represent the entire Alexander terrane.

The chronology of microplate movements in southeastern Alaska, suggested by paleomagnetism, can be summarized as follows. Between the late Carboniferous and the Late Triassic the Craig subterrane arrived at its present latitude relative to the craton after moving approximately 15° north [*Van der voo et al.*, 1980]. Possibly, this part of the Alexander terrane originated near present-day northern California, a hypothesis consistent with the paleomagnetic evidence. This hypothesis is supported by geologic similarities between the Alexander terrane and Paleozoic rocks of the Klamath Mountains [*Jones et al.*, 1972]. Northward drift of the neighboring Admiralty subterrane has not been established, so that the two blocks may have different translational histories before Permian time. The Craig and Admiralty subterranes were linked together sometime before deposition of the Lower Permian Pybus Formation, so any northward drift after that time would have affected both crustal blocks.

Paleomagnetic results from the Hound Island Volcanics suggest that northward drift of the Alexander terrane was complete by Late Triassic time. The basalt and limestone sequences of Wrangellia were deposited just before this time, somewhere far to the south near the paleoequator [*Hillhouse*, 1977*b*]. Then Wrangellia, after drifting approximately 27° north, arrived at its present latitude relative to the Alexander terrane and the craton. The final coalescence of border terranes in the northern Cordillera possibly occurred along subduction zones, where the continent converged with oceanic plates carrying fragments torn from island arcs or continents. Alternatively, the fragments may have been juxtaposed along right-lateral transform faults. Recent movements on faults, such as the fault in Chatham Strait, have obscured some of the original suture zones. In addition, the suture in Peril Strait between Wrangellia and the Craig subterrane is obscured by late Mesozoic intrusive rocks.

CONCLUSIONS

The Late Triassic pillow basalt and massive flows that border Keku Strait indicate a paleomagnetic pole at latitude 23.0°N, longitude 188.8°E (α_{95} = 11.0°). All specimens were partially demagnetized in AF's ranging from 15 to 40 mT. Vector diagrams obtained from stepwise demagnetization of selected specimens revealed that a single component of magnetization had been isolated in 75% of the specimens; the rest show evidence of a low-intensity high-coercivity component, usually directed along the recent geomagnetic field direction. Bias due to this secondary magnetization is considered to be negligible because the fold test was strongly positive. In addition, normal and reversed directions are antiparallel at the 95% confidence level. The stable magnetism is presumably carried by titanomagnetite that was magnetized during initial cooling of the basalts. Many magnetic grains consist of latticed titanomagnetite-ilmenite intergrowths, the products of high-temperature oxidation.

The paleolatitude determined from the Hound Island Volcanics does not differ significantly from the position predicted by paleomagnetic data from central North America. Therefore northward drift of the Alexander terrane, determined from previous paleomagnetic investigations of Paleozoic rocks, was complete by Late Triassic time. Northward drift of Wrangellia, the westernmost pre-Cretaceous terrane, occurred after the Triassic, so that Wrangellia and the Alexander terrane moved north independently. Apparently, the region surrounding northern Keku Strait was rotated (about a vertical axis) ~100° in the counterclockwise sense. This rotation was probably confined to a small block within the Alexander terrane, because older rocks on Prince of Wales Island [*Van der Voo et al.*, 1980] show a smaller counterclockwise rotation of 25°.

Acknowledgments. We thank L. J. P. Muffler for providing geologic information in advance of publication. W. D. Davis assisted in the field work, and E. A. Mankinen, K. Pagan, and G. Thrupp made most of the magnetic measurements.

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(Received June 20, 1979;
accepted January 7, 1980.)

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